

Quantitative methods and survey design

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Content of the lecture

1) Quantitative methods in digital media and consumer research

2) Principles and concepts of quantitative research

3) Designing survey questionnaire

4) Statistical procedures with SPSS

5) Usability scales and questionnaires

6) Free online tools



Survey on quantitative methods (5 min)

- Please answer this survey about your previous experience on quantitative methods:
- https://forms.gle/5B6DyxMwkQUYvmfU6
- All answers are anonymous
- Thank you for your answers!



Quantitative methods in consumer research

- Long tradition in market and consumer research
- Consumer behaviour, needs and attitudes towards consumption
- Customer satisfaction and loyalty
- Consumer trends
- Longitudinal changes over time

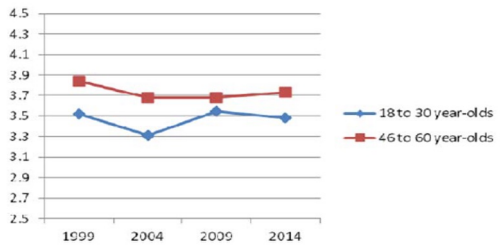


FIGURE 1 Ecological attitudes towards consumption among young adults and late middle-agers between 1999 and 2014 (1 = completely disagree, 5 = completely agree). [Colour figure can be viewed at wileyonlinelibrary.com]

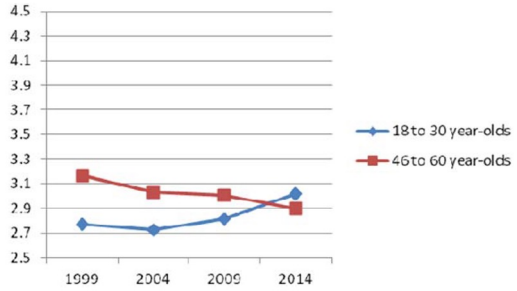


FIGURE 2 Economical attitudes towards consumption among young adults and late middle-agers between 1999 and 2014 (1 = completely disagree, 5 = completely agree). [Color figure can

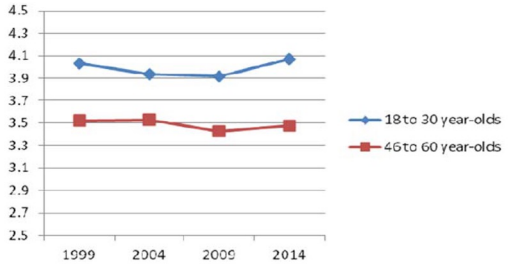


FIGURE 3 Self-indulgent attitudes towards consumption among young adults and late middle-agers between 1999 and 2014

Quantitative methods in consumer research



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ORIGINAL ARTICLE

International Journal of Consumer Studies WILEY

Fashion sensitive young consumers and fashion garment repair: Emotional connections to garments as a sustainability strategy

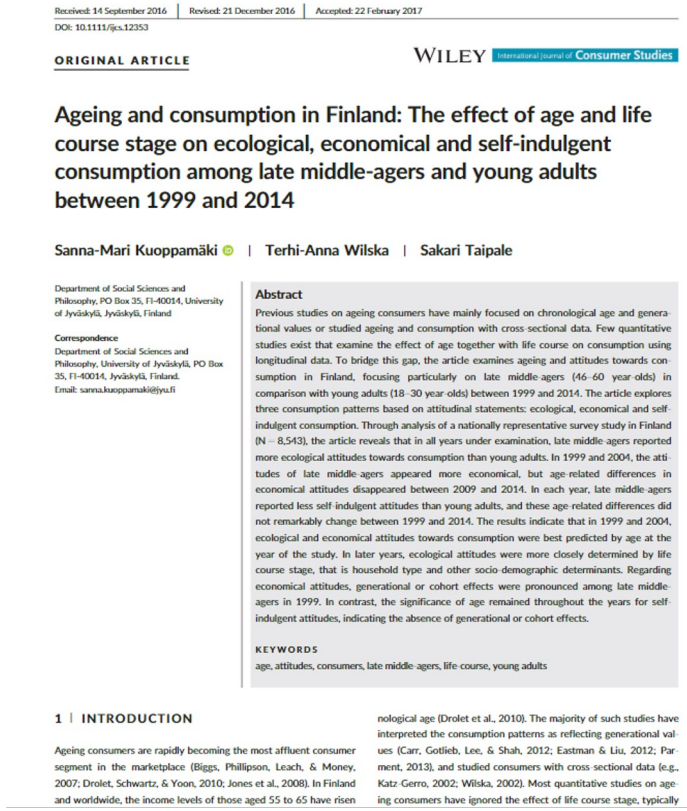
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Abstract
Where clothing consumption has continued to rise around the world, a deeper understanding of how and why garments are disposed of is critical in regard to addressing the issue of textile waste by consumers. The purpose of this study was to explore the garment management processes of young, fashion sensitive consumers, examining their disposal behaviours as well as motivations towards garment end-of-life extension through maintenance or repair of damaged fashion clothing. A survey of 161 South Korean young consumers (18–34 years) was conducted, utilizing a fashion sensitivity scale to measure impact on unsustainable garment disposal practices and garment repair behaviour. Further, general recycling behaviour of these individuals was examined, as linked to garment repair propensity. Results indicate that fashion sensitive consumers dispose of garments more rapidly, and with less ethical consideration. However, these same consumers are motivated to repair items that support their fashion identity. The study contributes some useful insight into encouraging garment life extension practise among heavy consumers of fashion, thus extending what is known about fashion garment disposal in the sustainability context.

KEYWORDS
consumer behaviour, fashion, recycling, repair, sustainable



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ORIGINAL ARTICLE

WILEY International Journal of Consumer Studies

Ageing and consumption in Finland: The effect of age and life course stage on ecological, economical and self-indulgent consumption among late middle-agers and young adults between 1999 and 2014

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Abstract
Previous studies on ageing consumers have mainly focused on chronological age and generational values or studied ageing and consumption with cross-sectional data. Few quantitative studies exist that examine the effect of age together with life course on consumption using longitudinal data. To bridge this gap, the article examines ageing and attitudes towards consumption in Finland, focusing particularly on late middle-agers (46–60 year-olds) in comparison with young adults (18–30 year-olds) between 1999 and 2014. The article explores three consumption patterns based on attitudinal statements: ecological, economical and self-indulgent consumption. Through analysis of a nationally representative survey study in Finland (N = 8,543), the article reveals that in all years under examination, late middle-agers reported more ecological attitudes towards consumption than young adults. In 1999 and 2004, the attitudes of late middle-agers appeared more economical, but age-related differences in economical attitudes disappeared between 2009 and 2014. In each year, late middle-agers reported less self-indulgent attitudes than young adults, and these age-related differences did not remarkably change between 1999 and 2014. The results indicate that in 1999 and 2004, ecological and economical attitudes towards consumption were best predicted by age at the year of the study. In later years, ecological attitudes were more closely determined by life course stage, that is household type and other socio-demographic determinants. Regarding economical attitudes, generational or cohort effects were pronounced among late middle-agers in 1999. In contrast, the significance of age remained throughout the years for self-indulgent attitudes, indicating the absence of generational or cohort effects.

KEYWORDS
age, attitudes, consumers, late middle-agers, life-course, young adults

1 | INTRODUCTION

Ageing consumers are rapidly becoming the most affluent consumer segment in the marketplace (Biggs, Phillipson, Leach, & Money, 2007; Drolet, Schwartz, & Yoon, 2010; Jones et al., 2008). In Finland and worldwide, the income levels of those aged 55 to 65 have risen

nological age (Drolet et al., 2010). The majority of such studies have interpreted the consumption patterns as reflecting generational values (Carr, Gotlieb, Lee, & Shah, 2012; Eastman & Liu, 2012; Parment, 2013), and studied consumers with cross-sectional data (e.g., Katz-Gerro, 2002; Wilska, 2002). Most quantitative studies on ageing consumers have ignored the effect of life course stage, typically

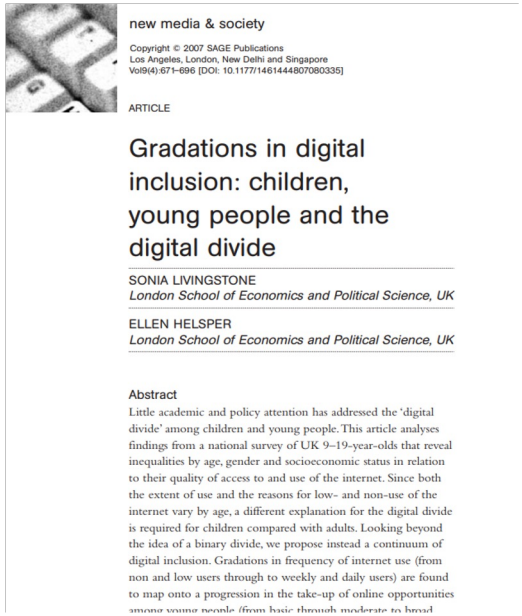
Quantitative methods in digital media research

- Long tradition in the median and communication studies
- Digital media use and well-being
- Mobile phone use and usage patterns in different cultural contexts
- Children, young adults and older adults as users of new media
- Digital inequalities
- ‘Digital detox’
- Digital inclusion - Social inclusion





Quantitative methods in digital media research



new media & society

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ARTICLE

Gradations in digital inclusion: children, young people and the digital divide

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Abstract
Little academic and policy attention has addressed the 'digital divide' among children and young people. This article analyses findings from a national survey of UK 9–19-year-olds that reveal inequalities by age, gender and socioeconomic status in relation to their quality of access to and use of the internet. Since both the extent of use and the reasons for low- and non-use of the internet vary by age, a different explanation for the digital divide is required for children compared with adults. Looking beyond the idea of a binary divide, we propose instead a continuum of digital inclusion. Gradations in frequency of internet use (from non and low users through to weekly and daily users) are found to map onto a progression in the take-up of online opportunities among young people (from basic through moderate to broad

Article

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Domestication of smartphones and mobile applications: A quantitative mixed-method study

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Abstract
Smartphones are finding their way into our daily lives. This paper examines the domestication of smartphones by looking at how the way we use mobile applications affects our everyday routines. Data is collected through an innovative quantitative mixed-method approach, combining log data from smartphones and survey (perception)



Article

The emergence of a "digital underclass" in Great Britain and Sweden: Changing reasons for digital exclusion

new media & society
2017, Vol. 19(8) 1253–1270
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DOI: 10.1177/1461444816634676
journals.sagepub.com/home/nms
SAGE

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Abstract
Research into reasons for Internet non-use has been mostly based on one-off cohort studies and focused on single-country contexts. This article shows that motivations for being offline changed between 2005 and 2013 among non- and ex-users in two high-diffusion European countries. Analyses of Swedish and British data demonstrate that non-user populations have become more concentrated in vulnerable groups.

Exploring data ageism through statistics

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Article

Exploring data ageism: What good data can('t) tell us about the digital practices of older people?

Mireia Fernández-Ardèvol [1] **and Line Grenier** [2]

Abstract
Considering that data are no stranger to politics and power, we argue that it may well be a site of age-based discrimination. We discuss how older people are described and, at times, disregarded in the analysis of digitisation and how those partial descriptions bring about challenges in the study of digital practices throughout life. We propose the notion of data ageism to conceptualise the production and reproduction of the disadvantaged status of old age caused by decisions concerning how to collect and deliver whose data. We exemplify this concept by examining data produced by Eurostat, the statistical office of the European Union, which offers high-quality statistics on digitisation, but no data on individuals aged 75 years and over.

Keywords
Ageism, data, data ageism, digital life, old age, statistics

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- Using data produced by Eurostat to explore ageism in data
- Eurostat: Statistical office in European Union offers high quality data on digitalisation, but no data on indiviuals aged 75+

→What is absent in data?

→Who does the data represents?

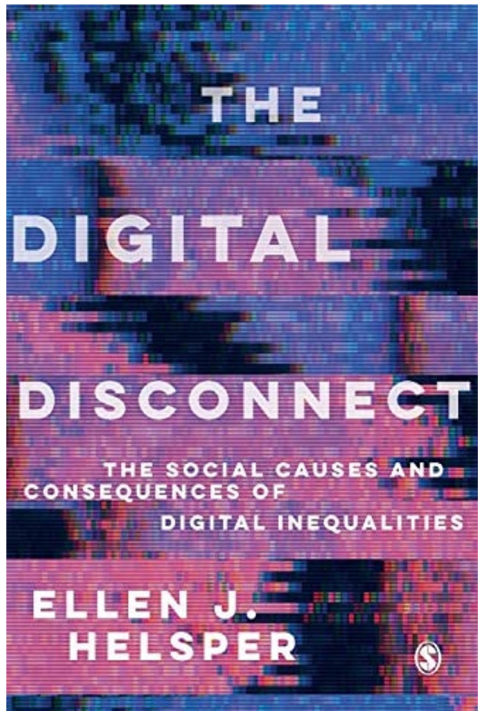
→ How are the measurements in data designed?

Discussion on preparatory readings (5 min)

- Based on your preparatory readings…

- *What are the measurement techniques to investigate digital inclusion/exlusion?*

- *What are the most important socio-demographic variables that explain digital inclusion?*



Measuring digital inclusion

Van Deursen and Van Dijk (2018)

- *First level* of digital divide: Access to digital technology
- *Second level* of digital divide: Skills to digital technology
- *Third level* of digital divide: Benefits from digital technology

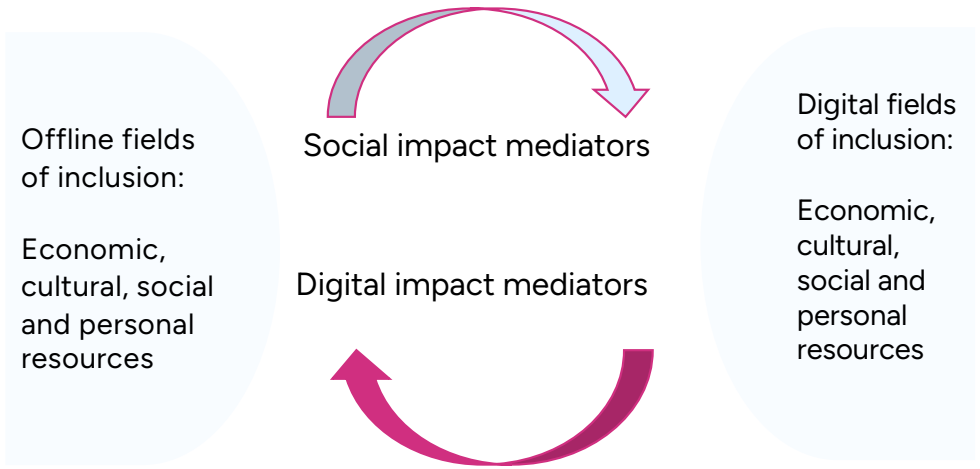


- Shift from"Access/no access – Use/non-use" to different benefits and different patterns of use"
- The concept of 'Warm expert' (Bakardjieva, 2005)
- Socio-demographic variables (age, education and income level) have traditinoally been the most strongest predictors of digital technology use
- The significance of gender in technology use?
- "Digital natives vs. digital immigrants (Prensky, 2001) has grown old"

Corresponding fields model of digital and social inclusion (Helsper, 2012)

The connection between digital and social inclusion is reciprocal

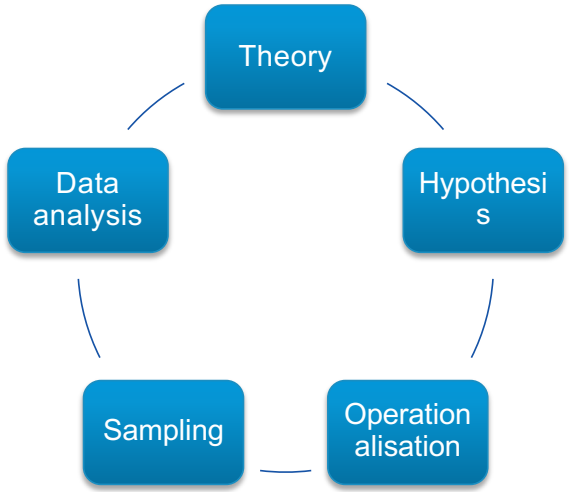
- Digital inclusion: Level of engagement with digital technologies
- Social inclusion: Experiences of connectivity, lack of isolation and loneliness



Principles of quantitative research

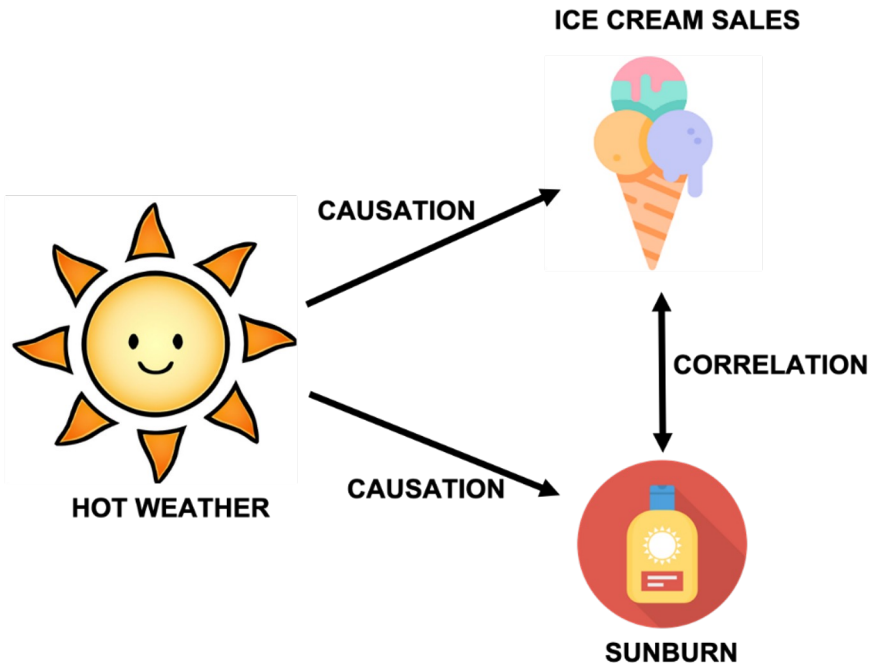
- Collection of numerical data
- Investigating frequencies, prevalences, and creating generalisations
- Causality or correlation between two or more variables

- Deductive logic: Deriving hypothesis from theory
- Operationalisation: turning abstracts concepts into measurable estimate
- Indicator: Indirect measure of a concept (e.g., income is an indicator for socio-economic background)

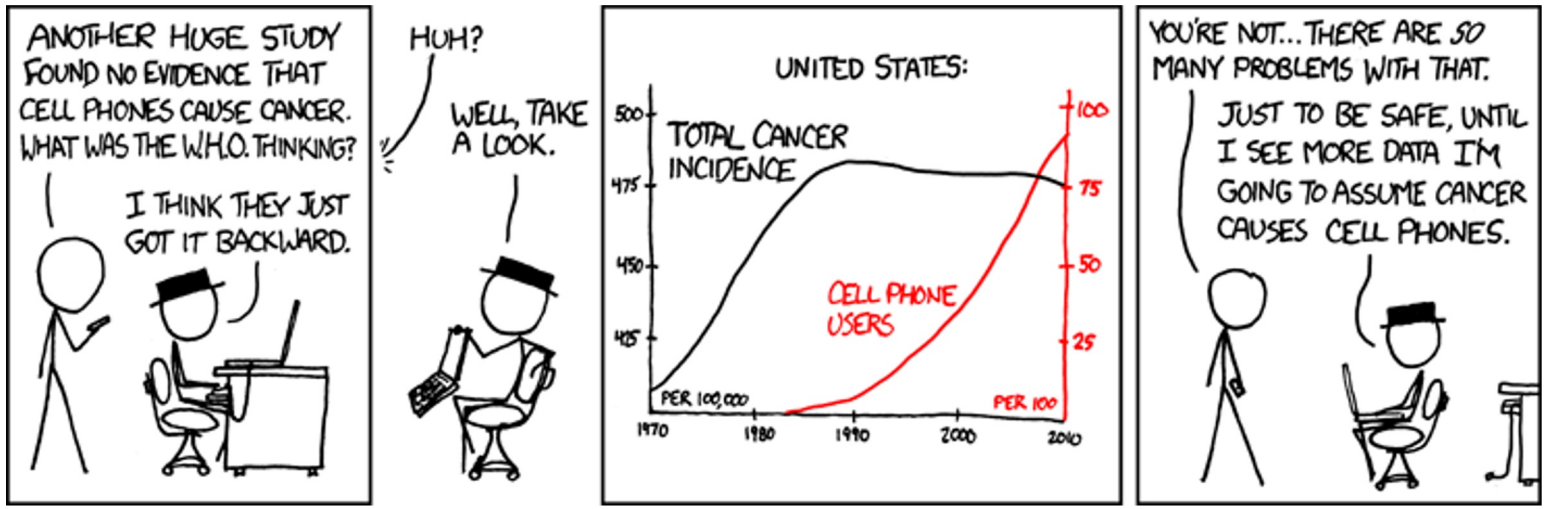


Causation vs. correlation

There might be a correlation between ice cream and sunburn, but not a causal relation.



Causation vs. correlation



Validity, reliability and statistical significance

- Assessment of *trustworthiness* of the research and procedure
- *Reliability*: Consistency of a measure. The measurement produces similar results over time.
- *Validity:* The concept really measure what is should measure
- Examples of the reliability or validity

- Statistical significance: Whether or not the observed difference is accurate or due to change alone (null hypothesis)
- *Remember: Statistical significance shows the accuracy of the difference, but it does not tell about the significance (meaningfulness) of the difference as such!*

Statistical significance and P-value

- There is always a possibility that an observed effect would have occurred due to sampling error
- Null hypothesis: Two variables are not related in the population (e.g. there is no relationship between gender and visiting the gym)
- Statistical tests are used to assess the statistical significance are p-value and chi-square test
- If the *p*-value of an observed effect or difference is less than (or equal to) 0.05, a researcher may conclude that the effect reflects the characteristics of the whole population, thereby rejecting the null hypothesis.
- P-value should be between 0.00 and 0.05 to show a statistically significant difference

Chi-square test and Cronbach's alfa

Chi-Square test

- The chi-square ($\chi 2$) test is applied to contingency tables. It allows us to establish how confident we can be that there is a relationship between the two variables in the population.
- The test works by calculating for each cell in the table an expected frequency or value
- The chi-square value means nothing on its own and can be interpreted only in relation to its associated level of statistical significance

Cronbach's Alfa

- Cronbach's alfa is a measure of internal consistency. It is the most common test score reliability coefficient. Applied for instance in factor analysis or evaluation of the reliability of Likert scale.

Example of the Chi-square test

reasons for visiting * gender Crosstabulation

			gender		
			Male	Female	Total
reasons for visiting	relaxation	Count	3	6	9
		% within gender	7.1%	12.5%	10.0%
	fitness	Count	15	16	31
		% within gender	35.7%	33.3%	34.4%
	lose weight	Count	8	25	33
		% within gender	19.0%	52.1%	36.7%
	build strength	Count	16	1	17
		% within gender	38.1%	2.1%	18.9%
Total		Count	42	48	90
		% within gender	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	22.726[a]	3	.000
Likelihood Ratio	25.805	3	.000
Linear-by-Linear Association	9.716	1	.002
N of Valid Cases	90		

a. 2 cells (25.0%) have expected count less than 5. The minimum expected count is 4.20.

Interpret the Pearson Chi-Square values for information about chi-square

Bryman, A. (2012). Social research methods. 4th edition.

Chapter 16. Using IBM SPSS for Windows

Factor analysis and Cronbach's Alfa

TABLE 1 Attitudes towards ecological, economical and self-indulgent consumption, factor analysis (1999–2014)

	Ecological	Economical	Self-indulgent
Statement			
'I am worried about the environmental effects of my consumption'	.709		
'The world is overloaded with goods and life is too consumption-oriented'	.343		
'I consciously prefer ecological products'	.707		
'I am worried about the origins and health risks of food that is sold to consumers'	.472		
'Everyone should save money for a rainy day'		.589	
'In my opinion, my lifestyle is economical'		.532	
'I will fund my purchases by saving in advance'		.557	
'I am concerned about what other people think of me'			.414
'I take good care of my looks'			.404
'I want to derive pleasure from my consumption'			.506
'I often eat out'			.390
'I often shop impulsively'			.421
Eigenvalue, % (cumulative %)	21.3 (21.3)	14.1 (35.3)	10.2 (46.0)
Cronbach's Alfa	.645	.567	.525

Break 10 min



Sampling and survey research

- Researcher needs to formulate research questions
- Researcher needs to decide what kind of population is suited to the investigation
- Developing a questionnaire: Operationalisation of abstract concepts
- Sampling
  - Population based sample
  - Panel
  - Sample size
- Cross-sectional survey vs. longitudinal survey
- Response rate





How to design a survey questionnaire

- Formulate your research questions based on previous research and theory
- Formulate clear hypotheses
- Questionnaire should have different sections and questionnaire patterns
- ”What is the association between age, technology use and well-being?”
  - Socio-demographic variables (age, gender, education, income, household type)
  - Questions measuring technology use (frequence, time, place, device)
  - Questions measuring well-being or quality of life
- Example: ACT project





Usability scales

- How to assess the usability of a new mobile health application for behavioural change?

- System Usability Scale (SUS)
- Service User Technology Acceptability Questionnaire (SUTAQ)
- Patient Activation Measure
- Mobile Health App Usability Questionnaire (MAUQ)
- Telehealth Usability Questionnaire
- Chatbot usability questionnaire



System Usability Scale (SUS)

Participants are asked to score the following 10 items with a Likert Scale from 1 to 5 (1= Strongly disagree 5= Strongly agree)

- I think that I would like to use this system frequently.
- I found the system unnecessarily complex.
- I thought the system was easy to use.
- I think that I would need the support of a technical person to be able to use this system.
- I found the various functions in this system were well integrated.
- I thought there was too much inconsistency in this system.
- I would imagine that most people would learn to use this system very quickly.
- I found the system very cumbersome to use.
- I felt very confident using the system.
- I needed to learn a lot of things before I could get going with this system.

Service User Technology Acceptance Questionnaire

- Service User Technology Acceptability Questionnaire (SUTAQ) is a questionnaire tool to measure patients' perception and acceptability of Telehealth products.

- Torbjørnsen, A., Småstuen, M. C., Jenum, A. K., Årsand, E., & Ribu, L. (2018). The Service User Technology Acceptability Questionnaire: Psychometric Evaluation of the Norwegian Version. *JMIR human factors*, *5*(4), e10255. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6324518/

Mobile health app usability questionnaire

- Mobile Health App Usability Questionnaire (MAUQ) is explicitly designed to measure the usability for Medical Health Applications.
- 21 questions where these questions are divided into three different subsections.

    1. Section. Eight Questions measuring ease of use and satisfaction.

    2. Section. Six questions measuring System information arrangement.

    3. Section. Seven questions measuring Usefulness

Zhou L, Bao J, Setiawan IMA, Saptono A, Parmanto B
The mHealth App Usability Questionnaire (MAUQ): Development and Validation Study. JMIR Mhealth Uhealth 2019;7(4):e11500. https://mhealth.jmir.org/2019/4/e11500

Chatbot usability questionaire (CUQ)

- The chatbot's personality was realistic and engaging
- The chatbot seemed too robotic
- The chatbot was welcoming during initial setup
- The chatbot seemed very unfriendly
- The chatbot explained its scope and purpose well
- The chatbot gave no indication as to its purpose
- The chatbot was easy to navigate
- It would be easy to get confused when using the chatbot
- The chatbot failed to recognise a lot of my inputs
- Chatbot responses were useful, appropriate and informative
- Chatbot responses were not relevant
- he chatbot coped well with any errors or mistakes
- The chatbot seemed unable to handle any errors
- he chatbot was very easy to use

- 16 statements measuring the usability of chatbots

- https://www.ulster.ac.uk/research/topic/computer-science/artificial-intelligence/projects/cuq

Reflect with your pair

- What are the benefits and challenges of survey methodologies in user studies?
- Think about usability vs. user experiece
- How to capture the difference?



Statistical analysis with SPSS

- Bryman (2014): Chapter 16
  - Defining variables
  - Recoding variables
  - Computing new variables
  - Generating a contingency table

- Tutorial for Factor analysis: http://www.open.ac.uk/socialsciences/spsstutorial/files/tutorials/factor-analysis.pdf
- Video tutorial for Linear Regression (London School of Economics): https://www.youtube.com/watch?v=4UxSfuX8DjM

Free online tools for survey design

- Google Forms: www.google.com/forms
- SurveyMonkey: https://www.surveymonkey.com/
- 1ka: https://www.1ka.si/en







What's next?

- Your user study should include a selection and description of method to study users + plan for data collection
- Reflection in small groups (10 min):
  - What are the research questions in your user study?
  - How would you operationalise them?
  - How would you design the questionnaire? What sections to include?
  - How would you create the sample?
- Use Padlet: https://padlet.com/sannaku/Bookmarks
- Answer the same questions in your short reaction paper!

